

Some 3-regular 4-ordered graphs *

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Abstract

A graph G is k -ordered if for any sequence of k distinct vertices $v_1, v_2, \dots, v_k$ of G there exists a cycle in G containing these k vertices in the specified order. In 1997, Ng and Schultz posed the question of the existence of 3-regular 4-ordered graphs other than K_4 and $K_{3,3}$. In 2008, Meszaros solve the question by proving the Petersen graph and the Heawood graph are 3-regular 4-ordered graphs. Moreover, the generalized Honeycomb torus $\text{GHT}(3, n, 1)$ is 4-ordered for any even integer n with $n \geq 8$. Up to now, all the known 3-regular 4-ordered graphs are vertex transitive. Among these graphs, there are only two non-bipartite graphs, namely the complete graph K_4 and the Petersen graph. In this paper, we prove that there exists a bipartite non-vertex-transitive 4-ordered 3-regular graphs of order n for any sufficient large even integer n . *Keywords:* k -ordered; cubic graphs

1 Introduction

For the graph definitions and notation, we follow [?]. Let $G = (V, E)$ be a graph if V is a finite set and E is a subset of $\{(u, v) \mid (u, v) \text{ is an unordered pair of } V\}$. We say that V is the *vertex set* and E is the *edge set*. Two vertices u and v are *adjacent* if $(u, v) \in E$. A graph is of order n if $|V| = n$. The *degree* of a vertex u in G , denoted by $\deg_G(u)$, is the number of vertices adjacent to u . A graph is k -regular if $\deg(x) = k$

for any $x \in V(G)$. A *cubic graph* is a 3-regular graph. A *path* of length $m - 1$, $\langle v_0, v_1, \dots, v_{m-1} \rangle$, is an ordered list of distinct vertices such that v_i and v_{i+1} are adjacent for $0 \leq i \leq m - 2$. We also write the path $\langle v_0, \dots, v_k, P, v_l, \dots, v_m \rangle$ for $P = \langle v_k, \dots, v_l \rangle$. A *cycle* is a path with at least three vertices such that the first vertex is the same as the last one.

A graph G is k -ordered if for any sequence of k distinct vertices $v_1, v_2, \dots, v_k$ of G there exists a cycle in G containing these k vertices in the specified order. The concept of k -ordered graphs was introduced in 1997 by Ng and Schultz [?]. Previous results concerning cycle order ability focus on minimum degree and forbidden subgraph conditions that imply k -orderedness [?, ?, ?, ?]. A comprehensive survey of results can be found in [?].

In [?], Ng and Schultz posed the question of the existence of 3-regular 4-ordered graphs other than K_4 and $K_{3,3}$. In [?], Meszaros solve the question by proving the Petersen graph and the Heawood graph are 3-regular 4-ordered graphs. Moreover, the generalized Honeycomb torus $\text{GHT}(3, n, 1)$ is 4-ordered if n is an even integer with $n \geq 8$. Up to now, all the known 3-regular 4-ordered graphs are vertex transitive. Among these graphs, there are only two non-bipartite graphs, namely the complete graph K_4 and the Petersen graph. In this paper, we prove that there exists a bipartite non-vertex-transitive 3-regular 4-ordered graphs of order n for any sufficient large even integer n . Moreover, there exists a bipartite non-vertex-transitive 3-regular 4-ordered graphs of order n for any sufficient large even integer n .

In the following section, we give some exam-

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ples of 3-regular 4-ordered graphs and discuss their properties. In section 3, we introduce the concept of 4-ordered cells. Using 4-ordered cell, we can construct more 3-regular 4-ordered graphs. In the final section, we give some unsolved problems.

The following lemma proved in [?] will be used later.

Lemma 1.1. *Any 4-ordered 3-regular graph with more than 6 vertices does not contain a cycle of length 4.*

2 Generalized honeycomb torus

Stojmenovic [?] proposed three classes of honeycomb torus architectures: honeycomb hexagonal torus, honeycomb rectangular torus, and honeycomb rhombic torus. Cho and Hsu [?] found that all these honeycomb torus networks can be characterized in a unified way, and thereby proposed a class of interconnection networks known as the generalized honeycomb torus.

Let n be a positive even integer with $n \geq 4$, m be a positive integer, and d be a nonnegative integer that is less than n and is of the same parity as m . An (m, n, d) generalized honeycomb torus, denoted by $GHT(m, n, d)$, is a graph with the vertex set $\{(i, j) \mid i \in \{0, 1, \dots, m-1\}, j \in \{0, 1, \dots, n-1\}\}$. We call m , n , and d as the *width*, *height*, and *slope* of $GHT(m, n, d)$, respectively. For a vertex (i, j) of $GHT(m, n, d)$, i and j are called its *first* and *second components*, respectively. Here and in what follows, all arithmetic operations carried out on the first and second components are modulo m and n , respectively. Two vertices (i, j) and (k, l) with $i \leq k$ are adjacent if and only if one of the following three conditions is satisfied:

- (1) $(k, l) = (i, j + 1)$ or $(k, l) = (i, j - 1)$;
- (2) $0 \leq i \leq m - 2$, $i + j$ is odd, and $(k, l) = (i + 1, j)$;
- (3) $i = 0$, j is even, and $(k, l) = (m - 1, j + d)$.

The generalized honeycomb torus $GHT(3, 8, 1)$, $GHT(4, 8, 0)$, and $GHT(4, 8, 2)$, for example, are shown in Figure ???. Obviously, any $GHT(m, n, d)$ is 3-regular and vertex transitive. We can label those vertices (i, j) white when $i + j$ is even or

black if $i + j$ is odd. This makes $GHT(m, n, d)$ bipartite. It is proved in [?] that $\{GHT(m, n, d) \mid m \text{ is even and } d = 0\}$ is the set of honeycomb rectangular tori. In [?], an infinite family of 3-regular 4-ordered graphs is proposed. Actually, this family of graphs is $\{GHT(3, n, 1) \mid n \text{ is an even integer with } n \geq 8\}$. Using our terminology, we prove the following lemma in [?].

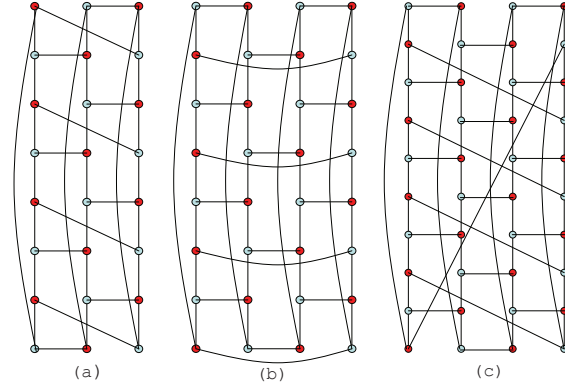


Figure 1: The generalized honeycomb torus (a) $GHT(3, 8, 1)$, (b) $GHT(4, 8, 0)$, and (c) $GHT(4, 10, 2)$.

Lemma 2.1. *Any $GHT(3, n, 1)$ is 4-ordered for any even n with $n \geq 8$.*

Proof. We prove this lemma by induction. Using computer programming, we can check that $GHT(3, 8, 1)$ is 4-ordered. Assume that n is any positive even integer with $n \geq 10$ and assume that $GHT(3, n - 2, 1)$ is 4-ordered. Let x_1, x_2, x_3 , and x_4 be any four vertices of $GHT(3, n, 1)$. We want to find a cycle in $GHT(3, n, 1)$ that traverses the vertices x_1, x_2, x_3 , and x_4 by the indices. Since $n \geq 10$, there exists an integer j such that $\{x_1, x_2, x_3, x_4\} \cap \{(r, s) \mid r \in \{0, 1, 2\}, s \in \{j, j + 1\}\} = \emptyset$. In $GHT(3, n, 1)$, we delete all the vertices in $\{(r, s) \mid r \in \{0, 1, 2\}, s \in \{j, j + 1\}\}$ and join $(i, j - 1)$ with $(i, j + 2)$ for $i = 0, 1, 2$. See Figure ?? for illustration. Obviously, the resultant graph is isomorphic to $GHT(3, n - 2, 1)$. By assumption, there exists a cycle C' in $GHT(3, n - 2, 1)$ that traverses the vertices x_1, x_2, x_3 , and x_4 in that order. Now, we replace all the edges of the form joining $(i, j - 1)$ to $(i, j + 2)$ in C' with the path $\langle (i, j - 1), (i, j), (i, j + 1), (i, j + 2) \rangle$ to obtain a cycle C in $GHT(3, n, 1)$ that traverses the vertices x_1, x_2, x_3 , and x_4 by the indices. The lemma is proved. $\square$

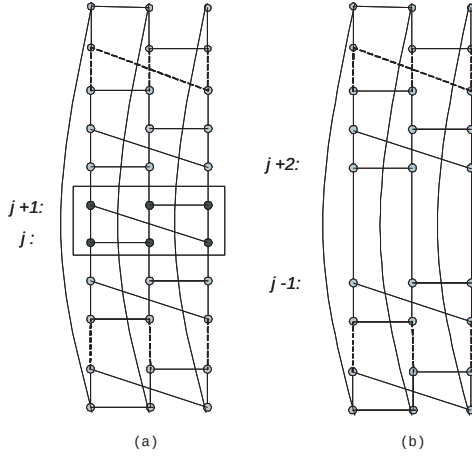


Figure 2: Illustration for the proof of Lemma ??: (a) $\text{GHT}(3, n, 1)$ and (b) delete all the vertices in $\{(r, s) \mid r \in \{0, 1, 2\}, s \in \{j, j+1\}\}$ and join $(i, j-1)$ with $(i, j+2)$ for $i = 0, 1, 2$.

Remarks. In the above proof, we have seen that the desired path pattern of $\text{GHT}(3, n, 1)$ can be obtained from a path pattern of $\text{GHT}(3, n-2, 1)$ by inserting two rows. We call this operation *row insertion*.

Theorem 2.1. Assume that m is an odd integer with $m \geq 3$ and n is an even integer with $n \geq 4$. The generalized honeycomb torus $\text{GHT}(m, n, 1)$ is 4-ordered if and only if $n \neq 4$.

Proof. Obviously, there are 4-cycles in $\text{GHT}(m, 4, 1)$. By Lemma ??, $\text{GHT}(m, 4, 1)$ is not 4-ordered. Using computer programming, we can check that $\text{GHT}(m, n, 1)$ is 4-ordered for $m \in \{3, 5, 7\}$ and $n \in \{6, 8\}$. Using row insertion, we can prove that $\text{GHT}(m, n, 1)$ is 4-ordered for $m \in \{3, 5, 7\}$ and $n \geq 10$.

Now, assume that $m \geq 9$ and assume that $\text{GHT}(m-2, n, 1)$ is 4-ordered. Let x_1, x_2, x_3 , and x_4 be any four vertices of $\text{GHT}(m, n, 1)$. We want to find a cycle in $\text{GHT}(m, n, 1)$ that traverses the vertices x_1, x_2, x_3 , and x_4 by the indices. Since $m \geq 10$, there exists an integer j such that $\{x_1, x_2, x_3, x_4\} \cap \{(r, s) \mid r \in \{0, 1, \dots, m-1\}, s \in \{0, 1, \dots, n-1\}\} = \emptyset$. In $\text{GHT}(m, n, 1)$, we delete all the vertices in $\{(r, s) \mid r \in \{0, 1, \dots, m-1\}, s \in \{0, 1, \dots, n-1\}\}$ and join $(i-1, j)$ with $(i+2, j)$ for $j \in \{0, 1, \dots, n-1\}$ if (i, j) is adjacent to $(i, j+1)$. See Figure ?? for illustration. Obviously, the resultant graph is isomorphic to $\text{GHT}(m-2, n, 1)$. By assumption, there exists a cycle C' in $\text{GHT}(m-2, n, 1)$ that traverses the

replace all the edges of the form joining $(i-1, j)$ to $(i+2, j)$ in C' with the path $\langle (i-1, j), (i, j), (i, j+1), (i+1, j+1), (i, j+1), (i, j+2) \rangle$ to obtain a cycle C in $\text{GHT}(m, n, 1)$ that traverses the vertices x_1, x_2, x_3 , and x_4 by the indices. The lemma is proved. $\square$

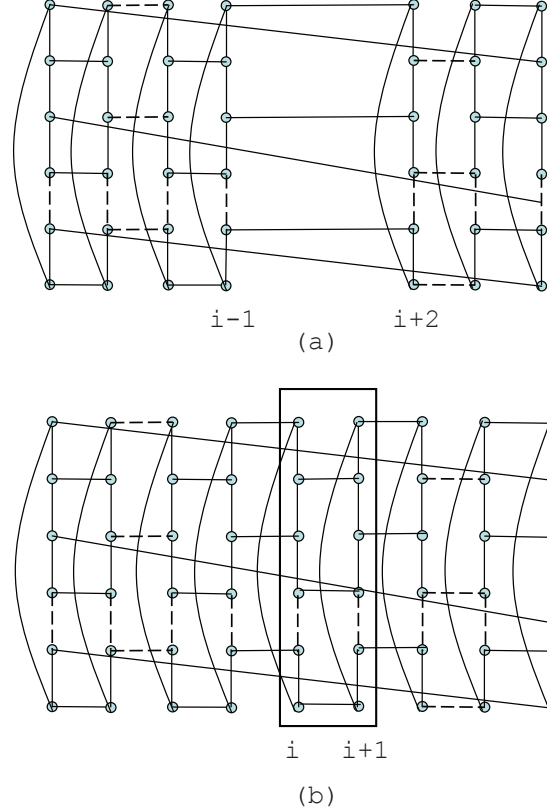


Figure 3: Illustration for the proof of Lemma ??: (a) $\text{GHT}(m, n, 1)$ and (b) delete all the vertices in $\{(r, s) \mid r \in \{0, 1, \dots, m-1\}, s \in \{0, 1, \dots, n-1\}\}$ and join $(i-1, j)$ with $(i+2, j)$ for $j \in \{0, 1, \dots, n-1\}$ if (i, j) is adjacent to $(i, j+1)$.

Remarks. In the above proof, we have seen that the desired path pattern of $\text{GHT}(m, n, 1)$ can be obtained from a path pattern of $\text{GHT}(m-2, n, 1)$ by inserting two columns. We call this operation *column insertion*.

Using similar techniques as above, we have the following theorem.

Theorem 2.2. Assume that m is a positive even integer with $m \geq 2$ and n is an even integer

with $n \geq 4$. The rectangular honeycomb torus $GHT(m, n, 0)$ is 4-ordered if and only if $n \neq 4$. The 27th Workshop on Combinatorial Mathematics and Computational Theory

Now, we discuss some path patterns for generalized honeycomb torus.

Let (a, b) be any edge of $GHT(m, n, d)$. We use $XHT^{a,b}(m, n, d)$ to denote the subgraph $GHT(m, n, d) - \{a, b\}$. Obviously, there are four vertices p, q, r , and s in $XHT^{a,b}(m, n, d)$ of degree 2 and all the other vertices are of degree 3.

Using computer programming, we can check that for any edge (a, b) of $GHT(m, n, 1)$ and any three vertices x_1, x_2 , and x_3 of $XHT^{a,b}(m, n, 1)$ with $m \in \{3, 5, 7\}$ and $n \in \{6, 8\}$ there exists a path P of $XHT^{a,b}(m, n, 1)$ joining u to v with $\{u, v\} \in \{p, q, r, s\}$ that traverses x_1, x_2 , and x_3 by the indices. Apply row insertion and column insertion, we can obtain the following lemma.

Lemma 2.2. Assume that m is odd with $m \geq 3$ and n is even with $n \geq 6$. Let (a, b) be any edge of $GHT(m, n, 1)$ and x_1, x_2 , and x_3 be three vertices of $XHT^{a,b}(m, n, 1)$. There exists a path P of $XHT^{a,b}(m, n, 1)$ joining u to v with $\{u, v\} \in \{p, q, r, s\}$ that traverses x_1, x_2 , and x_3 by the indices.

Using similar techniques, we have the following lemmas.

Lemma 2.3. Assume that m is odd with $m \geq 3$ and n is even with $n \geq 6$. Let (a, b) be any edge of $GHT(m, n, 1)$ and x be any vertex of $XHT^{a,b}(m, n, 1)$. There exists a path P of $XHT^{a,b}(m, n, 1)$ joining u to v with $\{u, v\} \in \{p, q, r, s\}$ that traverses x .

Lemma 2.4. Assume that m is odd with $m \geq 3$ and n is even with $n \geq 6$. Let (a, b) be any edge of $GHT(m, n, 1)$ and x_1 and x_2 be two vertices of $XHT^{a,b}(m, n, 1)$. Among all the twelve 2-permutations from $\{p, q, r, s\}$, there are at least seven 2-permutations u, v such that there exists a path P of $XHT^{a,b}(m, n, 1)$ joining u to v that traverses x_1 and x_2 by the indices.

Lemma 2.5. Assume that m is odd with $m \geq 3$ and n is even with $n \geq 6$. Let (a, b) be any edge of $GHT(m, n, 1)$. For any partition that divides $\{p, q, r, s\}$ into two pairs $\{\{u, v\}, \{w, x\}\}$, there exist two disjoint paths P and Q of $XHT^{a,b}(m, n, 1)$ such that P joins u to v and Q joins w to x .

Lemma 2.6. Assume that m is odd with $m \geq 3$ and n is even with $n \geq 6$. Let (a, b) be any

vertices of $XHT^{a,b}(m, n, 1)$. Among all the three partitions that divides $\{p, q, r, s\}$ into two pairs, there are at least two different $\{\{u, v\}, \{w, x\}\}$ such that there exist two disjoint paths P and Q of $XHT^{a,b}(m, n, 1)$ where P joins u to v traversing x_1 and Q joins w to x traversing x_3 .

3 4-ordered Cells

A cell is a 5-tuple (G, p, q, r, s) , where G is a graph, and p, q, r, s are four distinct vertices in G . It is cubic if $\deg_G(x) = 3$ for every $x \in V(G) - \{p, q, r, s\}$ and $\deg_G(x) = 2$ for $x \in \{p, q, r, s\}$. The graph $O_r((G, p, q, r, s))$ is obtained from G by adding two vertices a, b , and five edges (a, b) , (a, p) , (a, q) , (b, r) and (b, s) . We also say that the cell (G, p, q, r, s) is derived from $O_r((G, p, q, r, s))$ by deleting two adjacent vertices a and b .

A 4-ordered cell is a cell (G, p, q, r, s) with the following properties:

- (1) $O_r((G, p, q, r, s))$ is 4-ordered.
- (2) Let x_1, x_2 , and x_3 be any three vertices of G . There exists a path P of G that joining u to v with $\{u, v\} \in \{p, q, r, s\}$ that traverses x_1, x_2 , and x_3 by the indices.
- (3) Let x be any vertex of G and $\{u, v\}$ be any two vertices of $\{p, q, r, s\}$. There exists a path P of G joining u to v that traverses x .
- (4) Let x_1 and x_2 be two vertices of G . Among all the twelve 2-permutations from $\{p, q, r, s\}$, there are at least seven 2-permutations u, v such that there exists a path P of G joining u to v that traverses x_1 and x_2 by the indices.
- (5) For any partition that divides $\{p, q, r, s\}$ into two pairs $\{\{u, v\}, \{w, x\}\}$, there exist two disjoint paths P and Q of G such that P joins u to v and Q joins w to x .
- (6) Let x_1 and x_3 be two vertices of G . Among all the three partitions that divides $\{p, q, r, s\}$ into two pairs, there are at least two different $\{\{u, v\}, \{w, x\}\}$ such that there exist two disjoint paths P and Q of G where P joins u to v traversing x_1 and Q joins w to x traversing x_3 .

With the discussion in the previous section, we have the following theorem.

Theorem 3.1. *Every $XHT^{a,b}(m, n, 1)$ is a 4-ordered cell if m is odd with $m \geq 3$, n is even with $n \geq 6$ and (a, b) is any edge of $GHT(m, n, 1)$. Similarly, every $XHT^{a,b}(m, n, 0)$ is a 4-ordered cell if m is even with $m \geq 4$, n is even with $n \geq 6$ and (a, b) is any edge of $GHT(m, n, 1)$.*

By brute force, we can check the following theorem.

Theorem 3.2. *Let H be the Heawood graph and (a, b) be any edge of H . Then $H - \{a, b\}$ is a 4-ordered cell. However, let P be the Petersen graph and (a, b) be any edge of P . Then $P - \{a, b\}$ is not a 4-ordered cell.*

Cells are used for the construction of various families of graphs [?, ?, ?]. In this section, we will use the following operation to combine two cells. Let $C_i = (G_i, p_i, q_i, r_i, s_i)$ be a cell for $i = 1, 2$ and f be a 1-1 correspondence between $\{p_1, q_1, r_1, s_1\}$ and $\{p_2, q_2, r_2, s_2\}$. The graph $O_f(C_1, C_2)$ is obtained from the disjoint union of G_1 and G_2 by adding the edges $(p_1, f(p_1))$, $(q_1, f(q_1))$, $(r_1, f(r_1))$, and $(s_1, f(s_1))$. Obviously, all vertices in $O_f(C_1, C_2)$ are of degree 3 if C_1 and C_2 are cubic.

Theorem 3.3. *Assume that $C_i = (G_i, p_i, q_i, r_i, s_i)$ is a cubic 4-ordered cell for $i = 1, 2$. Then $O_f(C_1, C_2)$ is 4-ordered 3-regular.*

Proof. Let x_1, x_2, x_3 , and x_4 be any four vertices of $O_f(C_1, C_2)$. We need to find a cycle of $O_f(C_1, C_2)$ that traverses x_1, x_2, x_3 and x_4 by the indices. Without loss of generality, we have the following four cases.

Case 1: x_1, x_2, x_3 , and x_4 are all in G_1 . We assume that $O_r((G_1, p_1, q_1, r_1, s_1))$ is the graph obtained from G_1 by adding two vertices a_1 and b_1 together with the edges set $\{(p_1, a_1), (q_1, a_1), (r_1, b_1), (s_1, b_1), (a_1, b_1)\}$. By Property 1, there exists a cycle C in $O_r((G_1, p_1, q_1, r_1, s_1))$ that traverses x_1, x_2, x_3 and x_4 by the indices. Suppose that none of p_1, q_1, r_1 , and s_1 is in C . Obviously, C is also a cycle $O_f(C_1, C_2)$ that traverses x_1, x_2, x_3 , and x_4 by the indices. Suppose that some of p_1, q_1, r_1 , and s_1 are in C . Without loss of generality, we have the following three subcases:

Subcase 1.1: $\langle p_1, a_1, q_1 \rangle$ and $\langle r_1, b_1, s_1 \rangle$ are subpaths of C . By Property 5, there exist two disjoint paths Q_1 and Q_2 in G_2 such that Q_1 joins $f(p_1)$ to $f(q_1)$ and Q_2 joins $f(r_1)$ to $f(s_1)$.

With C , we replace $\langle p_1, a_1, q_1 \rangle$ and $\langle r_1, b_1, s_1 \rangle$ with the path $\langle r_1, f(r_1), Q_2, f(s_1), s_1 \rangle$ to obtain a cycle C' in $O_f(C_1, C_2)$. Obviously, C' traverses x_1, x_2, x_3 , and x_4 by the indices.

Subcase 1.2: $\langle p_1, a_1, q_1 \rangle$ is a subpath of C but $\langle r_1, b_1, s_1 \rangle$ is not a subpath of C . By Property 3, there exists a path Q in G_2 such that Q joins $f(p_1)$ to $f(q_1)$. With C , we replace $\langle p_1, a_1, q_1 \rangle$ with the path $\langle p_1, f(p_1), Q, f(q_1), q_1 \rangle$ to obtain a cycle C' in $O_f(C_1, C_2)$. Obviously, C' traverses x_1, x_2, x_3 , and x_4 by the indices.

Subcase 1.3: $\langle p_1, a_1, b_1, r_1 \rangle$ is a subpath of C . By Property 3, there exists a path Q in G_2 such that Q joins $f(p_1)$ to $f(r_1)$. With C , we replace $\langle p_1, a_1, b_1, r_1 \rangle$ with the path $\langle p_1, f(p_1), Q, f(r_1), r_1 \rangle$ to obtain a cycle C' in $O_f(C_1, C_2)$. Obviously, C' traverses x_1, x_2, x_3 , and x_4 by the indices.

Case 2: x_1, x_2 , and x_3 are in G_1 and x_4 is in G_2 . By Property 2, there exists path P of G_1 joining u to v for some $\{u, v\} \in \{p, q, r, s\}$ that traverses x_1, x_2 , and x_3 by the indices. By Property 3, there exists a path Q in G_2 such that Q joins $f(v)$ to $f(u)$ that traverses x_4 . We set C as $\langle u, P, v, f(v), Q, f(u), u \rangle$. Obviously, C traverses x_1, x_2, x_3 , and x_4 by the indices.

Case 3: x_1 and x_2 are in G_1 ; x_3 and x_4 are in G_2 . By Property 4, there are at least seven 2-permutations u, v among all the twelve 2-permutations from $\{p_1, q_1, r_1, s_1\}$ such that there exists a path P of G_1 joining u to v that traverses x_1 and x_2 by the indices. Similarly, there are at least seven 2-permutations u', v' among all the twelve 2-permutations from $\{p_2, q_2, r_2, s_2\}$ such that there exists a path Q of G_2 joining v' to u' that traverse x_3 and x_4 by the indices. By the pigeonhole principle, there exists a 2-permutation u, v such that there exists a path P of G_1 joining u to v that traverse x_1 and x_2 by the indices and there exists a path Q of G_2 joining $f(v)$ to $f(u)$ that traverse x_3 and x_4 by the indices. We set C as $\langle u, P, v, f(v), Q, f(u), u \rangle$. Obviously, C traverses x_1, x_2, x_3 , and x_4 by the indices.

Case 4: x_1 and x_3 are in G_1 ; x_2 and x_4 are in G_2 . By Property 6, there are at least two different $\{\{u, v\}, \{w, x\}\}$ among all the three partitions that divides $\{p_1, q_1, r_1, s_1\}$ into two pairs such that there exist two disjoint paths P_1 and P_3 of G where P_1 joins u to v traversing x_1 and P_3

joins w to x traversing x_3 . Similarly, there are at least two different $\{\{u', v'\}, \{w', x'\}\}$ among all the three partitions that divides $\{p_2, q_2, r_2, s_2\}$ into two pairs such that there exist two disjoint paths P_2 and P_4 of G where P_2 joins u' to v' traversing x_1 and P_4 joins w' to x' traversing x_4 . By the pigeonhole principle, there exists a partition $\{\{u, v\}, \{w, x\}\}$ such that P_1 joins u to v , P_2 joins $f(v)$ to $f(w)$, P_3 joins w to x , and P_4 joins $f(x)$ to $f(u)$. We set C as $\langle u, P_1, v, f(v), P_2, f(w), w, P_3, x, f(x), P_4, f(u), u \rangle$. Obviously, C traverses x_1, x_2, x_3 , and x_4 by the indices.

The following theorem is a consequence of the above theorem.

Theorem 3.4. *There exists a bipartite non-vertex-transitive 3-regular 4-ordered graphs of order n for any sufficient large even integer n .*

4 Concluding Remarks

In this paper, we prove that there exists a non-bipartite non-vertex-transitive 3-regular 4-ordered graphs of order n for any sufficient large even integer n . However, there are some unsolved problems. We have seen some generalized honeycomb torus that are 4-ordered and some are not. Thus, we would like to classify all generalized honeycomb torus that are 4-ordered. Similarly, we would like to classify all generalized Petersen graphs that are 4-ordered. Finally, we note that the complete graph K_4 , the complete bipartite graph $K_{3,3}$, and the Heawood graph are the only currently known 4-ordered 3-regular hamiltonian graphs. It would be interesting to investigate the existence of other 3-regular 4-ordered hamiltonian graphs.

References

- [1] G. Chen, R.J. Gould, F. Pfender, New conditions for ordered k -Hamiltonian graphs, *Ars Combin.* 70 (2003) 245–255.
- [2] H. Cho, L. Hsu, Generalized honeycomb torus, *Inform. Process. Lett.* 86 (4) (2003) 185–190.
- [3] V. Chvátal, Flip-flops in hypohamiltonian graphs, *Canadian Mathematical Bulletin*, *Congressus Numerantium* (1987), 298.

- [4] J.B. Collins, F.F. Schmeichel, New flip-flop constructions for hypohamiltonian graphs, *Discrete Mathematics*, 18, 193, 1977.
- [5] J.R. Faundree, R.J. Faundree, Forbidden subgraphs that imply k -ordered and k -ordered hamiltonian, *Discrete Math.* 243 (2002) 91–108.
- [6] J.R. Faundree, R.J. Gould, F. Pfender, A. Wolf, On k -ordered bipartite graphs, *Electron. J. Combin.* 10 (2003) R11.
- [7] R.J. Faundree, Survey of results on k -ordered graphs, *Discrete Math.* 229 (2001) 73–87.
- [8] L.H. Hsu, C.K. Lin, *Graph Theory and Interconnection Networks*, CRC Press, 2008.
- [9] K. Meszaros, On 3-regular 4-ordered graphs, *Discrete Math.* 308 (2008) 2149–2155.
- [10] L. Ng, M. Schultz, k -Ordered hamiltonian graphs, *J. Graph Theory* 24 (1997) 45–57.
- [11] I. Stojmenovic, Honeycomb networks: topological properties and communication algorithms, *IEEE Trans. Parallel Distributed Systems* 8 (10) (1997) 1036–1042.
- [12] C.H. Tsai, T.K. Li, Two construction schemes for cubic Hamiltonian 1-node-Hamiltonian graphs. *Math. Comput. Modelling* 48 (2008), 656–661.